Project overview

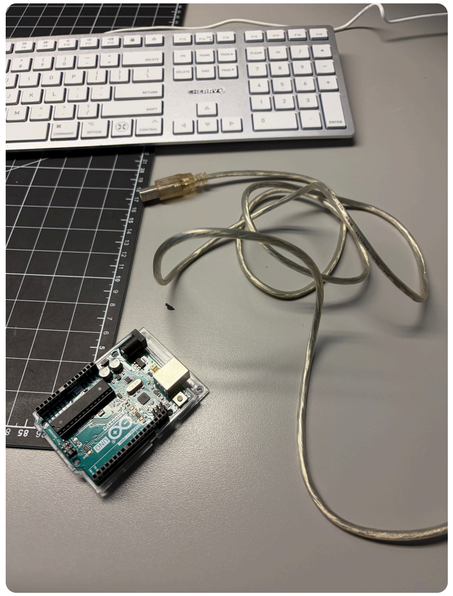
The class activities gave me a chance to try different hardware and software without having to settle on one idea right away. I was surprised by how many things could count as an input. After trying the examples and looking into responsive spaces, I became interested in an artwork that changes simply because someone is nearby

Design statement

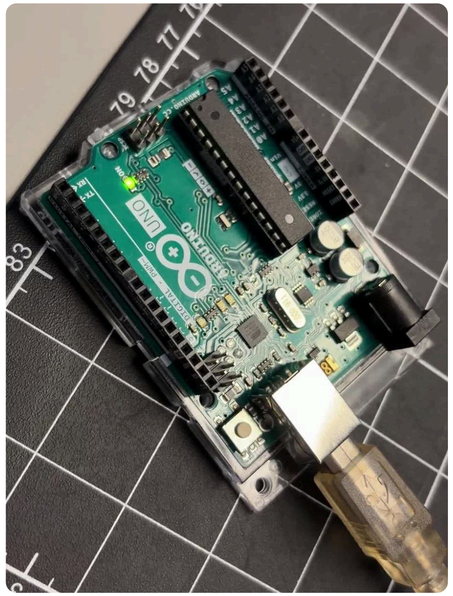
Still Here asks whether standing in one place can count as an interaction. In the proposed installation, a wall projection starts yellow and slowly shifts toward orange and red while someone stays in a marked area. When they leave, the image fades back. Most interfaces ask people to tap, swipe or move; I want to see what happens when simply staying changes the space. I like the idea of the room acknowledging someone without taking their picture or storing anything about them. I want the response to feel gradual, but not so subtle that someone mistakes it for a video playing on a loop. The artwork should be open to
interpretation, with enough feedback for a visitor to recognize that they caused the change. I am choosing the Art Path because the question of presence interests me more than building a conventional interface

Arduino: Input and output

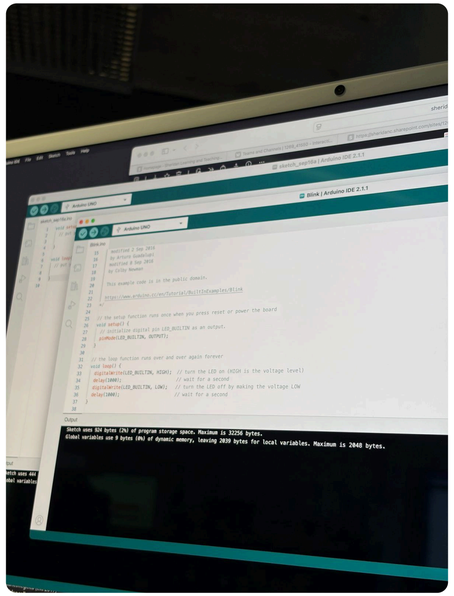
For the Arduino activity, I connected the Uno to the computer and worked through the provided Blink example in the Arduino IDE. Watching the LED switch on and off helped me understand what the code was doing



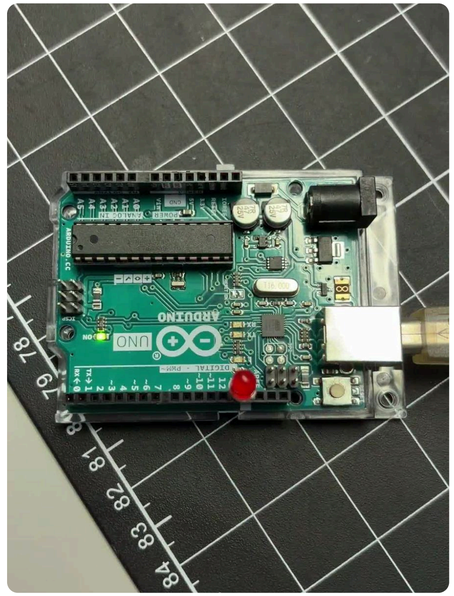
01 / Setup: Arduino Uno and USB cable before testing



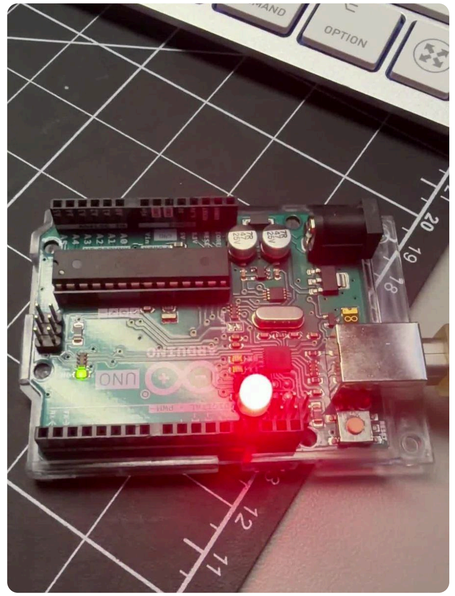
02 / Video still: first hardware activity



03 / Arduino IDE showing the Blink example



04 / Video still: continued Arduino exploration



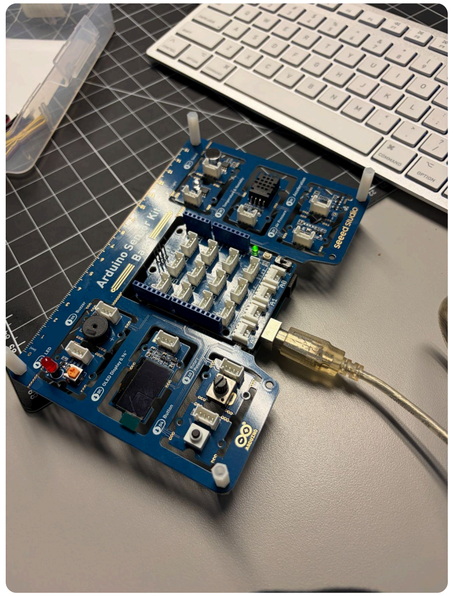
05 / Video still: further testing with LED output

Reflection

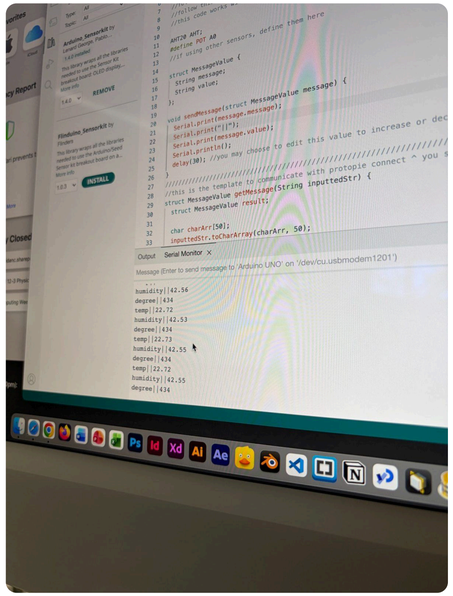
I liked testing the potentiometer because I could turn something physical and see the reading change immediately. It made me wonder what else that kind of input could control, beyond the LED and numbers we were looking at in class

ProtoPie Connect: Reading sensor values

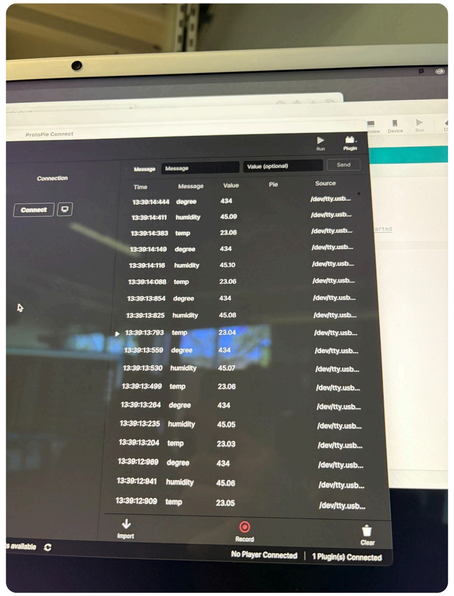
For ProtoPie Connect, followed the class setup for sending degree, temperature and humidity readings from the Arduino into the software. I worked through the connection and watched the values update. It was interesting to see the same physical data appear somewhere other than the Arduino IDE



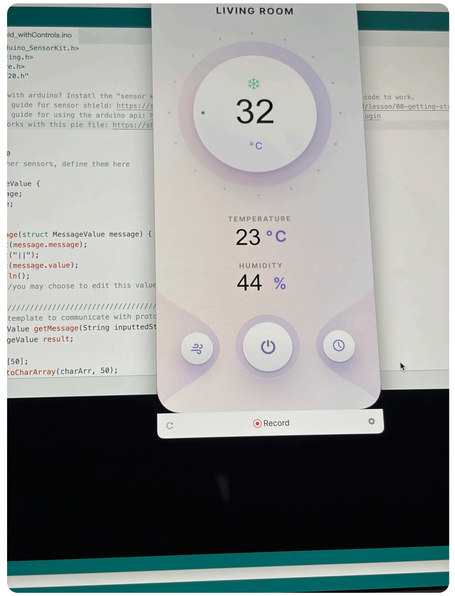
01 / Arduino hardware and sensor setup



02 / Arduino code and sensor readings



03 / Serial messages / incoming values



04 / ProtoPie Connect receiving messages

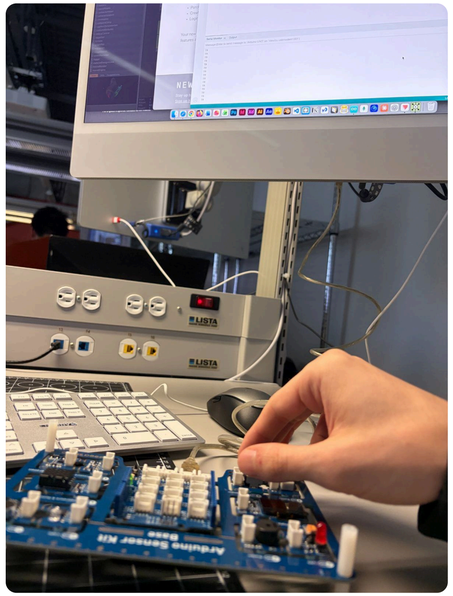
05 /

Reflection

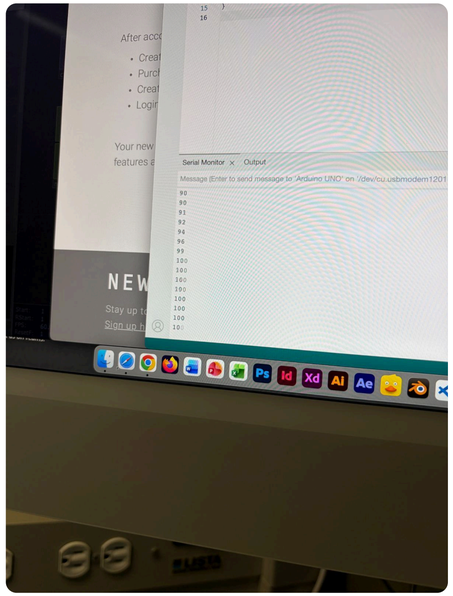
This activity made me think about how much information a sensor can collect without anyone tapping a screen. I started imagining other uses for those changing readings, like altering the look of an illustration or giving a room a visual response to its surroundings

TouchDesigner: Working with visuals

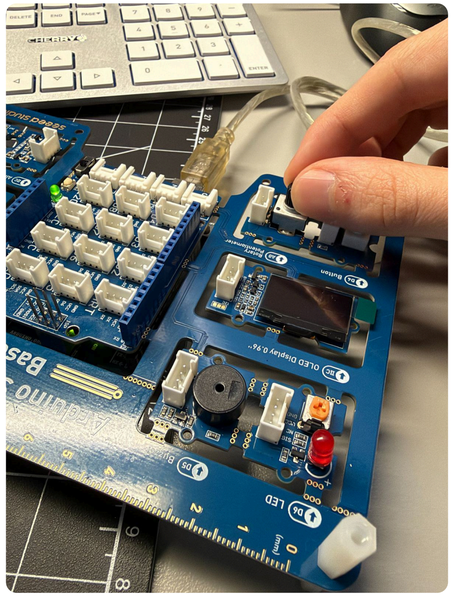
For TouchDesigner, used the Arduino connection and circular visual provided in the class example. I followed the setup and experimented with the input while looking at the graphic on screen. Out of the software we tried, this was the one that interested me most because I enjoy working with visuals



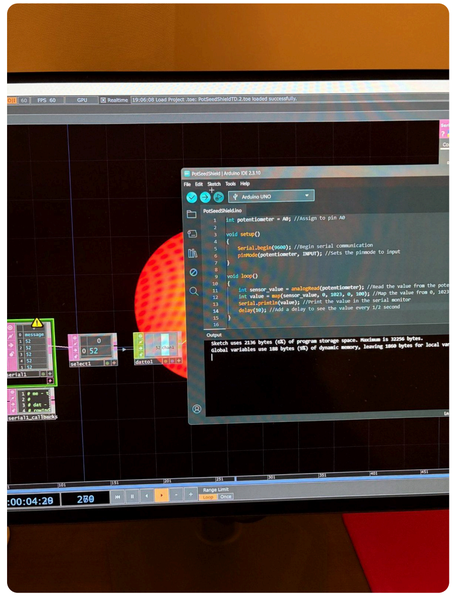
01 / Hardware connected during the activity



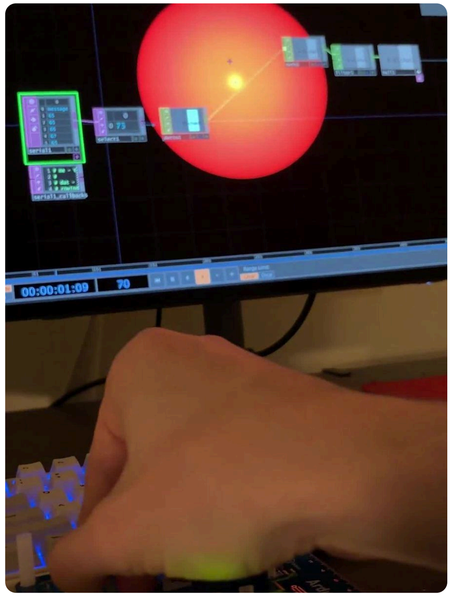
02 / Arduino sketch and software setup



03 / Serial data / input adjustment



04 / At-home session: code and circular graphic



05 / Video still: continued visual exploration

Reflection

TouchDesigner wouldn't run on the lab computer, so I brought the Arduino home and continued the exercise on my own computer. It was annoying to switch setups, but I was glad I could finish trying the example. Seeing how the image reacted got me thinking about other possibilities, including projections in a physical space. The direction that caught my attention

What I took from the activities

Tried the provided setups and I got a much better idea of what each tool could do. I liked having a physical input, but I was most interested in the way it could affect an image. That gave me a direction to explore, even though I didn't have a project picked out at the beginning

2027 - 2028 directions: responsive & touchless

I looked at two directions in home technology for 2027 - 2028: rooms that respond to the people in them, and controls that work without touching a screen. Both felt relevant to projection work, even though the studies are about consumer products rather than gallery visitors

Category 01 / Responsive environments

A responsive environment senses a change such as movement or presence and adjusts the surrounding conditions. Philips Hue's motion sensor demonstrates an established movement to light relationship; NIQ discusses smart home use and demand for connected experiences (NielsenIQ, 2026; Philips Hue, n.d.). In the detailed process research, 27% is reported as the share of Americans who used a smart-home product in the preceding 30 days. I treat this as a context statistic, not a measure of interactive art adoption

Category 02 / Touchless interaction

Touchless controls allow a physical gesture to become an input without a switch or screen. Google Nest Hub's Motion Sense illustrates gesture based controls; Worldpanel by Numerator reports ease of use as a criterion for 60% of surveyed U.S. smart-speaker buyers (Google, n.d.; Worldpanel by Numerator, 2026). That statistic does not demonstrate gesture demand, but it reinforces my goal of making feedback obvious.

Two possible visitors

I used Pew, YouGov and the smart-home research to imagine two people who might encounter the work. Nia and Alex are working personas rather than people I interviewed. I found the difference in their proposed expectations more useful than their ages

NIA / 26 / GEN Z

Goal hypothesis: be able to explore the work without reading lengthy instructions

Pew comparison: 33% of U.S. adults aged 18–29 reported owning a smart speaker with AI features

YouGov context: 49% of Gen Z Americans considered reviews important when choosing large appliances

ALEX / 40 / MILLENNIAL

Goal hypothesis: recognize the cause of a visual change almost immediately.

Pew comparison: 41% smart-speaker ownership with AI features among U.S. adults aged 30–49

YouGov context: 49% of U.S. Millennials considered appliance reviews important; the smart-home study also emphasizes ease of use

What the personas made me consider

Nia made me think about what somebody might try without instructions, while Alex made me think about how quickly a response becomes understandable. I wouldn't assume either behaviour from someone's age. The Pew and YouGov figures describe consumer habits, not the behaviour of people standing in front of an artwork (Pew Research Center, 2026; Bali, 2026)

Questions that shaped the idea

In the Week 5 lecture we looked at How Might I questions. I liked using them to name what could go wrong with an interaction before deciding on the exact hardware or visual effect (Rosala, 2021)

Responsive spaces

Problem 1	A slow-reacting projection could look like a prerecorded video if visitors do not know their presence matters
HMI 1	How might I make standing still produce a noticeable change without making it feel like an ordinary timer?
Problem 2	Motion-only sensors may fail to communicate that someone is still there
HMI 2	How might I show that remaining in one place is changing the image?
Problem 3	Motion-only sensors may fail to communicate that someone is still there
HMI 3	How might I show that remaining in one place is changing the image?

Touchless input

Problem 1	A visitor might not discover an invisible gesture interaction at all
HMI 1	How might I encourage visitors to discover a touchless interaction through the artwork itself?
Problem 2	If a gesture does not register, there may be no clear feedback
HMI 2	How might I make a small gesture produce understandable, playful visual feedback?

Choosing Art while retaining UX thinking

I enjoy UX/UI design, so I considered the Design Path too. I chose Art because I want to try something more open-ended this time, instead of building a tool with a specific function. I'd still use what I know from UX when thinking about whether visitors notice the interaction

Why the Art Path interests me

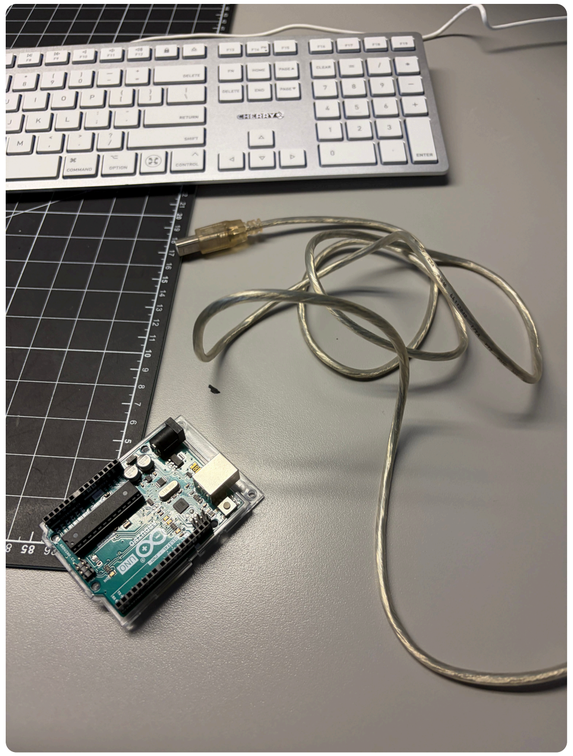
I could picture making a touchless museum interface, but that idea felt too much like invisible buttons. A projection that changes because someone is there has more room for experimentation. I like being able to think about stillness and time, not only whether a control is efficient

Why I will start individually

I want to begin on my own with one input and one projected result. That gives me enough room to change the timing and visuals as I learn what works. My first idea, which would connect two visitors through projected lines, is still an idea I like, but I would rather understand one person sensing first

Four visual directions

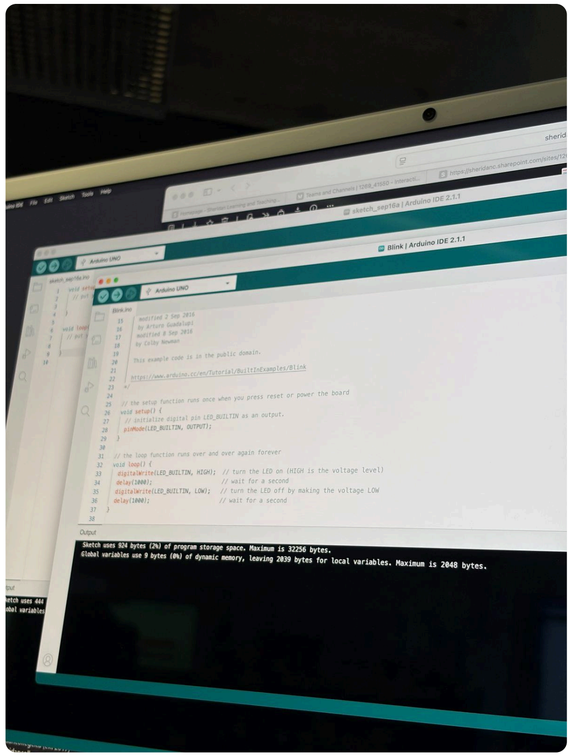
After the exercises and research, I considered four possible directions. I put them next to one another to compare the kinds of interaction I might enjoy developing



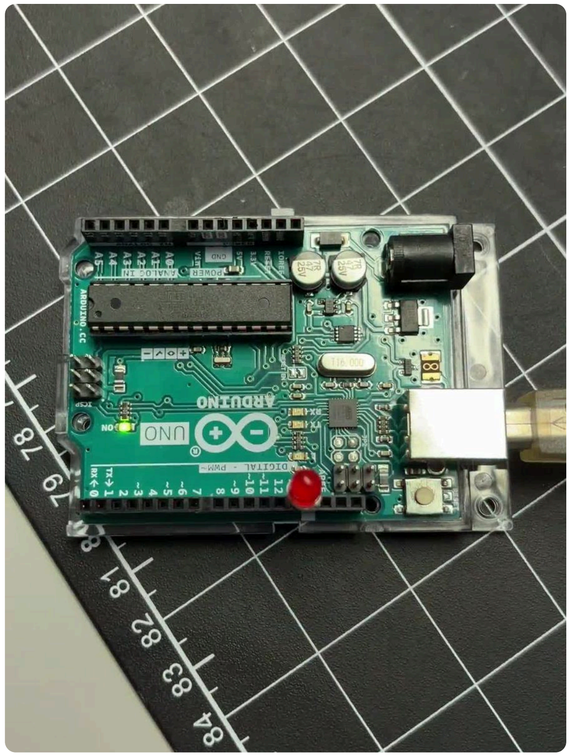
01 / STILL HERE: presence accumulated over time



02 / THE SOUND OF HOME: sound amplitude and abstract forms



03 / AFTERIMAGE: a lingering visual trace



04 / WEATHER YOU CAN'T HOLD: gestures as wind

Still Here: making stillness visible

With Still Here, I'm interested in what would happen if a projection changed while someone simply stayed in one place. There would be no button to press or gesture to learn



presence accumulated over time

Interaction narrative

Arrival. A person steps into a defined zone. The projection begins as yellow and responds subtly to show it has registered their presence

Staying. The longer the person remains, the more the projection shifts toward orange and red. The image slowly makes elapsed time perceptible

Leaving. When the visitor leaves, the signal changes and the projected colour returns or gradually fades back to yellow

What interests me is the possibility that someone can change a space without performing for it. I am not fully convinced by the yellow-to-red palette yet; red could feel like an alarm or a warning that someone has stayed too long. I want the colour to build in intensity without making the visitor feel like they should leave. I'm not entirely sure the yellow-to-red palette is right. Red could make it seem like the visitor is doing something wrong by staying too long, so I want to explore how the colour actually feels when it changes slowly.

How I would start testing

The first thing I'd try is a simple presence input connected to Arduino, with TouchDesigner using that reading to change a projected colour. I want to know whether the effect feels responsive before developing the visuals any further

Stage	Proposed function
Input	Someone enters or leaves the active area
Sensor	A webcam, Kinect, or another occupancy sensor provides a presence signal
Visual	TouchDesigner changes the projection gradually over time

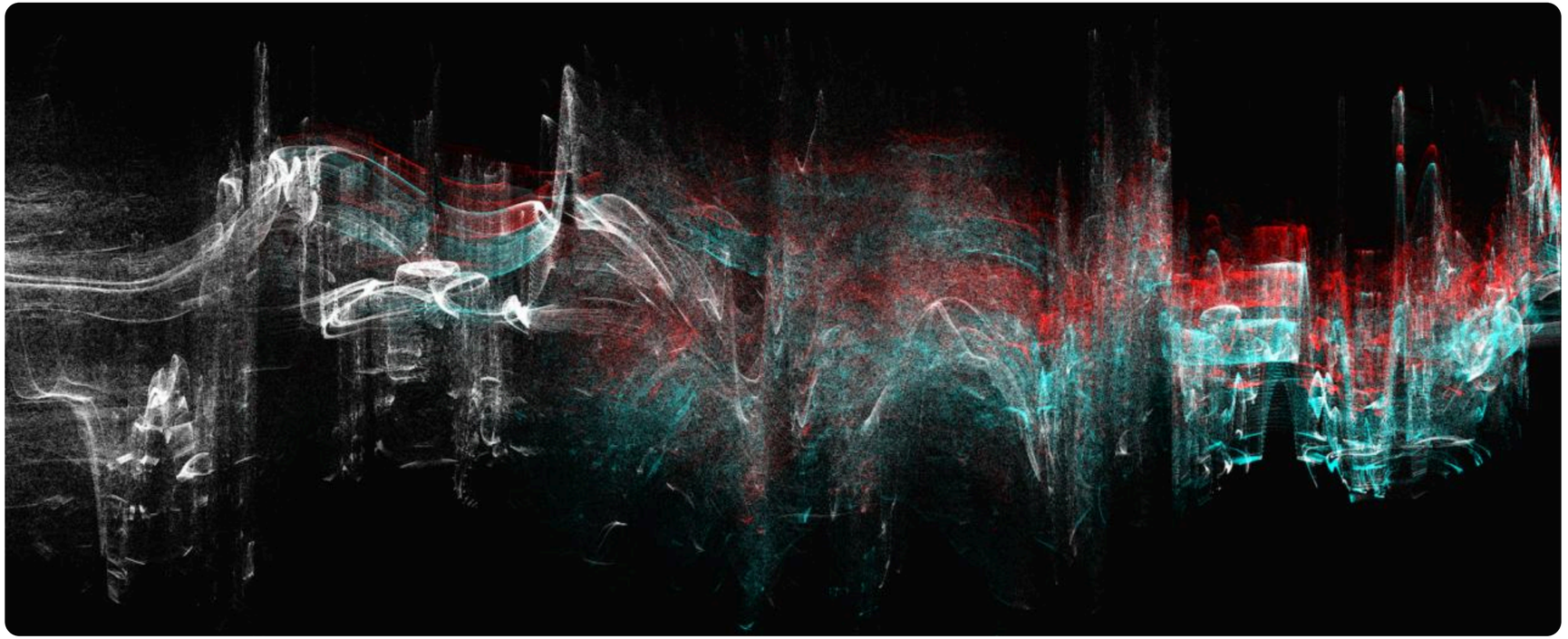
Questions for the first test

A motion-triggered sensor might miss someone once they stop moving. A webcam using person-detection software or a Kinect could be worth testing because the interaction depends on continued presence, not just movement. I also want to experiment with the timing; if the colour changes too slowly, someone might think it's a video playing on its own. I'd begin by checking whether the chosen setup can keep recognizing someone who stands still. Then I'd compare a quick visual acknowledgement with a slower colour shift, and ask someone to try the projection without an explanation to see what they notice

The Sound of (Home) (Or something else)

The Sound of Home was another idea I wanted to explore because it uses sounds already present in a room, rather than asking visitors to make a specific movement

https://derivative.ca/workshop/getting-started-touchdesigner/62938

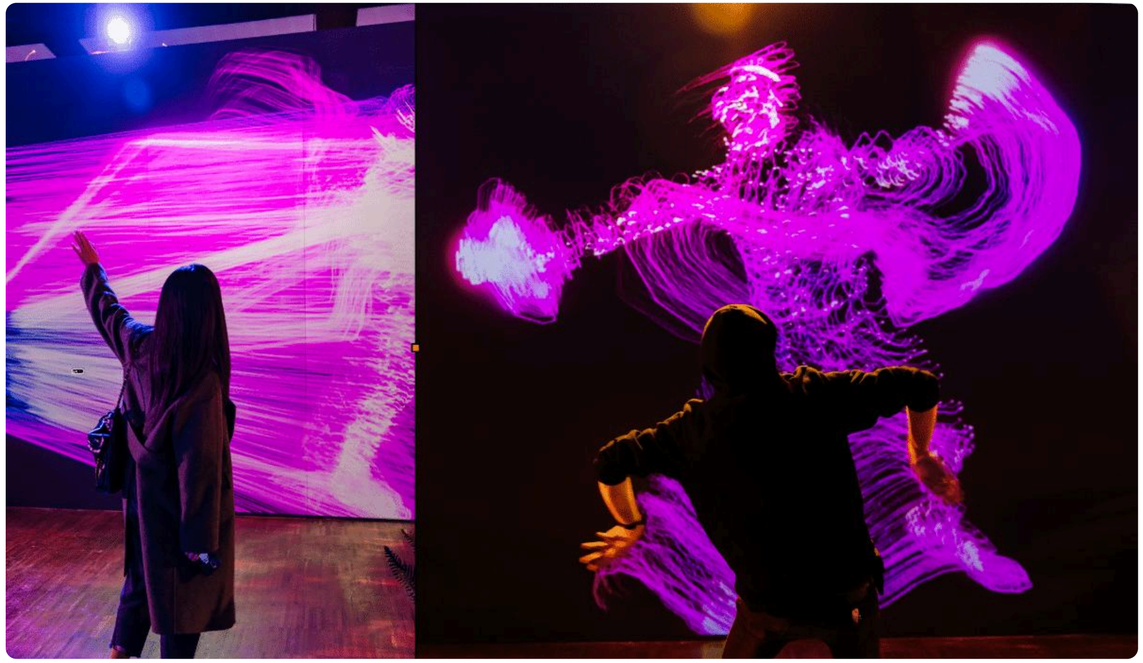


In this idea, a live microphone signal would change an abstract projection. Loudness could affect its movement or size, while quieter moments could leave the image almost still. I first imagined more than one room, but one microphone would be a more realistic place to begin

Afterimage & Weather You Can’t Hold

The Sound of Home was another idea I wanted to explore because it uses sounds already present in a room, rather than asking visitors to make a specific movement

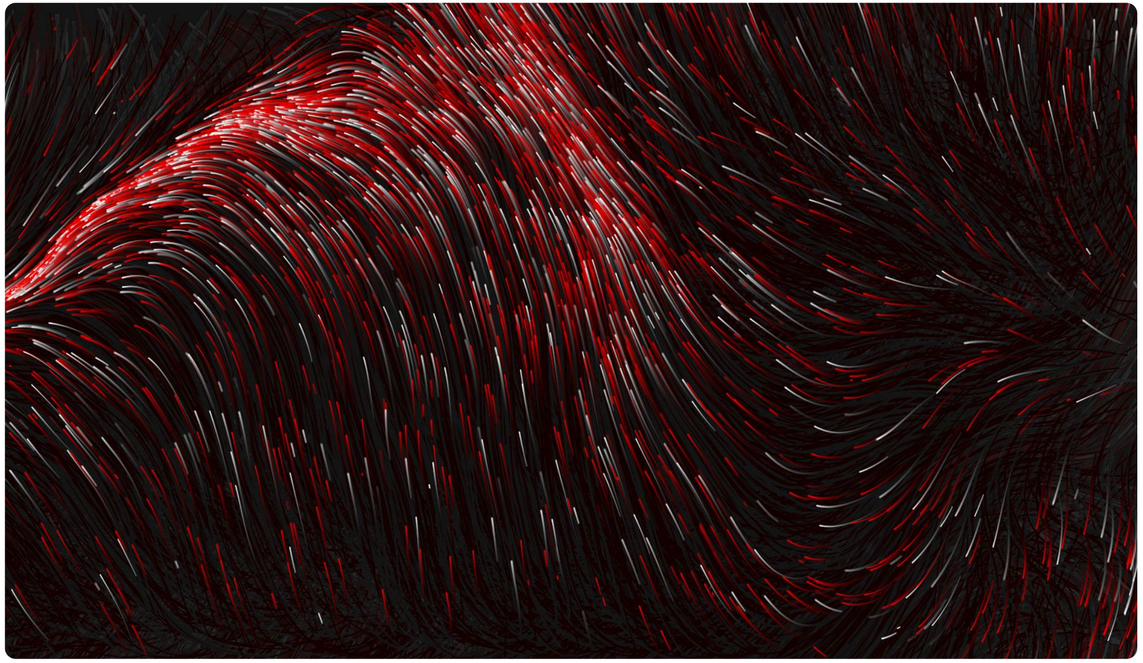
https://go2productions.com/blog/how-motion-tracking-enhances-interactive-installations/



Afterimage

Someone approaching a wall would leave thin projected marks that fade after they walk away. I like the idea of a space briefly keeping a trace of a visitor

https://forum.derivative.ca/t/particle-flow-workflow/112734



Weather You Can’t Hold

Small hand swipes would push projected particles like wind. It would be more playful and direct, though it depends on people discovering how the gestures work

For now, Still Here holds my attention more than the gesture idea because the visitor doesn't have to do much. I want to find out whether that can still make an interesting interaction

Thinking beyond a screen

The home-tech sources explained ways sensors already affect everyday spaces. I also looked for an artistic connection, because I don't want the result to feel like a smart-home demo

Immersive experience as context

Gensler Research Institute and Immersive Experience Institute (2025) discusses immersive experiences in which the environment is part of what people encounter. That interests me more than placing a typical interface on a wall

Taking Up Space: Isabelle Pleno

Isabelle Pleno's Taking Up Space (2024) gave me another reference for thinking about bodies and occupied space. My idea would use the time someone remains in a spot to affect the projection rather than relying on a button or a screen

Reflection and direction

I went into these activities mainly wanting to try the different tools and see what interested me. Since we were following examples provided in class, I wasn't creating the designs from scratch, but working through them gave me ideas about other ways the technology could be used. I didn't have a specific project in mind at the beginning.

Arduino and ProtoPie helped me understand how physical inputs could be read and used in different programs. TouchDesigner stood out the most because I enjoyed seeing the inputs affect something visual. It was frustrating when the lab computer wouldn't run the software, but continuing the activity at home gave me more time to explore it. I realized I was more interested in creating an interactive visual experience than building a traditional interface.

After looking into different project ideas, I started leaning toward Still Here. I like the idea of a space responding to someone simply being there, without requiring them to press anything or make a specific gesture. I'm still figuring out which sensor would work best and whether the visual changes would be noticeable enough. I think testing those things will help me understand what works and what I might need to change. These activities helped me find a direction I'm interested in exploring further, even though the concept is still developing

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